Chapter 4. Classification: Logistic Regression, Softmax and SVM

Logistic regression, multiclass classification, support vector machines

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1 Odds, Logit and the Logistic Model

Motivation. For a binary label we need a model of a probability, a number in $[0,1]$, as a function of features. Modelling the log-odds as linear gives the logistic model, whose output is the sigmoid of a linear score.

For a probability p, the odds are $o = p/(1-p)$ ($p = 0.8$ gives odds 4) and

$$\operatorname{logit}(p) = \log \frac{p}{1-p} \in \mathbb{R}, \qquad \sigma(z) = \frac{1}{1+e^{-z}} = \operatorname{logit}^{-1}(z).$$

Definition 4.1: logistic regression

With augmented input $\mathbf{x} = (1, x_1, \ldots, x_d)$ and $y \in \{0,1\}$,

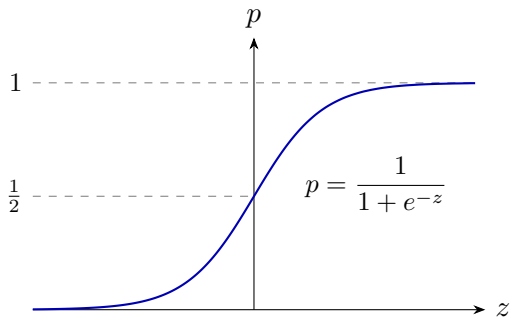
$$\log \frac{P(y=1 \mid \mathbf{x})}{P(y=0 \mid \mathbf{x})} = \mathbf{w}^\top \mathbf{x}, \qquad P(y=1 \mid \mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x}).$$

The score is linear and the probability transform nonlinear. Thresholding the probability at τ gives the hyperplane $\mathbf{w}^\top \mathbf{x} = \log \frac{\tau}{1-\tau}$ (at $\tau = 0.5$: $\mathbf{w}^\top \mathbf{x} = 0$).

Theorem 4.2: sigmoid identities

$\sigma'(z) = \sigma(z)(1-\sigma(z)) \le \frac{1}{4}$, $1 - \sigma(z) = \sigma(-z)$, $\frac{d}{dz}\log(1+e^z) = \sigma(z)$.

Proof. $\sigma'(z) = e^{-z}/(1+e^{-z})^2 = \sigma(z)\cdot\frac{e^{-z}}{1+e^{-z}}$ and $\frac{e^{-z}}{1+e^{-z}} = \frac{1}{e^z+1} = \sigma(-z) = 1-\sigma(z)$. The maximum of $s(1-s)$ on $[0,1]$ is $1/4$ at $s = 1/2$. ■



The logistic function, inverse of the logit $z = \log \frac{p}{1-p}$: log-odds z are mapped to probabilities p.

2 Likelihood, Gradient, Hessian and Convexity

Motivation. Logistic regression has no closed form, so we must optimize numerically. The Hessian is positive semidefinite, so the loss is convex: every local minimum is global and gradient methods are reliable.

For data $(\mathbf{x}_i, y_i)$ with $p_i = \sigma(\mathbf{w}^\top \mathbf{x}_i)$, $p(y_i \mid \mathbf{x}_i, \mathbf{w}) = p_i^{y_i}(1-p_i)^{1-y_i}$ and

$$J(\mathbf{w}) = -\sum_i \left[y_i \log p_i + (1-y_i)\log(1-p_i)\right] = \sum_i \left[\log(1+e^{z_i}) - y_i z_i\right], \quad z_i = \mathbf{w}^\top \mathbf{x}_i.$$

Theorem 4.3: logistic loss is convex

$\nabla J = \sum_i (p_i - y_i)\mathbf{x}_i = X^\top(\mathbf{p}-\mathbf{y})$ and $\nabla^2 J = \sum_i p_i(1-p_i)\mathbf{x}_i\mathbf{x}_i^\top = X^\top S X$, $S = \mathrm{diag}(p_i(1-p_i))$. Hence J is convex, and strictly convex if X has full column rank.

Proof. $\partial J/\partial z_i = \sigma(z_i) - y_i = p_i - y_i$ by the identities above, and $\partial z_i/\partial \mathbf{w} = \mathbf{x}_i$. Differentiating again, $\nabla p_i = p_i(1-p_i)\mathbf{x}_i$. For any $\mathbf{v}$, $\mathbf{v}^\top X^\top S X \mathbf{v} = \sum_i p_i(1-p_i)(\mathbf{x}_i^\top \mathbf{v})^2 \geq 0$; since $p_i(1-p_i) > 0$, the sum is positive whenever $X\mathbf{v} \neq \mathbf{0}$. ■

Example 4.4: one gradient step

$\mathbf{x} = (1,2)$, $y = 1$, $\mathbf{w} = \mathbf{0}$: $p = 1/2$, $\nabla J = (p-1)\mathbf{x} = (-1/2, -1)$. With $\eta = 0.1$, $\mathbf{w} = (0.05, 0.1)$, score 0.25, $p \approx 0.562$: the probability of the correct label increased.

3 Labels in $\{\pm 1\}$, Logistic Loss and BCE

Motivation. The same model can be written with labels $\{0,1\}$ (binary cross-entropy) or $\{\pm 1\}$ (logistic loss of the margin yf). The margin form connects logistic regression with the SVM and the other surrogate losses of this section.

With $Y \in \{\pm 1\}$ and score $f(\mathbf{x}) = \mathbf{a}^\top \mathbf{x} + b$, $P(Y = +1 \mid \mathbf{x}) = \sigma(f)$ and $P(Y = -1 \mid \mathbf{x}) = \sigma(-f)$, so $-\log P(Y \mid \mathbf{x}) = \ell(Yf)$ with the **logistic loss** $\ell(z) = \log(1 + e^{-z})$. $Yf(\mathbf{x})$ is the *margin*: positive iff the sign of the score is correct.

Theorem 4.5: three names for one objective

Let $P(-1) = (1,0)$, $P(+1) = (0,1)$, $\mu(r) = (\sigma(-r), \sigma(r))$, $\tilde{Y} = (1+Y)/2 \in \{0,1\}$ and $\mathrm{BCE}(s,t) = -[t \log s + (1-t)\log(1-s)]$. For every $(\mathbf{a}, b)$ and every i,

$$D_{\mathrm{KL}}(P(Y_i) \,\|\, \mu(f(\mathbf{x}_i))) = \mathrm{BCE}\left(\sigma(f(\mathbf{x}_i)), \tilde{Y}_i\right) = \ell(Y_i f(\mathbf{x}_i)).$$

So KL minimization, BCE minimization and logistic regression are the same problem.

Proof. For a one-hot first argument, KL $= -\log$(model probability of the true class). If $Y = +1$ this is $-\log \sigma(r) = \log(1+e^{-r}) = \ell(r)$, and BCE with $t = 1$ is $-\log s$. If $Y = -1$ it is $-\log \sigma(-r) = \ell(-r)$, and BCE with $t = 0$ is $-\log(1-s) = -\log \sigma(-r)$. ■

Theorem 4.6: ℓ is convex

$\ell'(z) = \sigma(z) - 1$ and $\ell''(z) = \sigma(z)\sigma(-z) > 0$.

3.1 Separable data

Theorem 4.7: no MLE on separable data

Suppose $Y_i(\mathbf{a}^{\star\top} X_i + b^\star) > 0$ for all i. Then $\inf_{\mathbf{a},b} \sum_i \ell(Y_i(\mathbf{a}^\top X_i + b)) = 0$, the infimum is not attained, and every minimizing sequence is unbounded. With an ℓ_2 penalty $\frac{\lambda}{2}(\|\mathbf{a}\|^2 + b^2)$ a unique minimizer exists and satisfies $\|\mathbf{a}_\lambda\|^2 + b_\lambda^2 \leq 2N \log 2/\lambda$.

Proof. Let $t = \min_i Y_i(\mathbf{a}^{\star\top} X_i + b^\star) > 0$ (finitely many positive numbers). Scaling by $C > 0$ and using that ℓ is decreasing, $L(C\mathbf{a}^\star, Cb^\star) \leq N \log(1 + e^{-Ct}) \to 0$; and $L \geq 0$. Every term

is strictly positive, so $L > 0$ everywhere. If a minimizing sequence were bounded, a convergent subsequence would give a point with $L = 0$. With the penalty: $L_\lambda \geq \frac{\lambda}{2}\|(\mathbf{a}, b)\|^2$ makes the sublevel set $\{L_\lambda \leq L_\lambda(\mathbf{0})\}$ compact, so a minimizer exists (Weierstrass); strong convexity gives uniqueness; and $\frac{\lambda}{2}\|(\mathbf{a}_\lambda, b_\lambda)\|^2 \leq L_\lambda(\mathbf{0}) = N\ell(0) = N\log 2$. ■

3.2 Surrogate risk bounds

For $(X, Y) \sim P$ and classifier $f = \operatorname{sign}(g)$, counting $g = 0$ as an error, the event $\{f(X) \neq Y\}$ implies $Yg(X) \leq 0$. For a nonnegative, nonincreasing loss ϕ this gives $\phi(Yg) \geq \phi(0)$ on errors and therefore

$$P(f(X) \neq Y) \leq \frac{\mathbb{E}[\phi(Yg(X))]}{\phi(0)}.$$

Logistic: $\phi(0) = \log 2$, so $R_{\log} < \varepsilon \Rightarrow P(f(X) = Y) > 1 - \varepsilon/\log 2$. Hinge $\max(0, 1-u)$ and exponential e^{-u} have $\phi(0) = 1$. Minimizing a convex surrogate thus controls the 0–1 error, which itself cannot be optimized by gradients.

4 Multiclass Classification and Softmax

Motivation. With K classes the sigmoid generalizes to the softmax, which turns K scores into a probability vector. Its cross-entropy loss has the same clean gradient (prediction minus target) as in the binary case.

For Q classes, $W \in \mathbb{R}^{Q\times d}$, scores $s_k = \mathbf{w}_k^\top \mathbf{x}$ and

$$p_k(\mathbf{x}) = \operatorname{softmax}(\mathbf{s})_k = \frac{e^{s_k}}{\sum_r e^{s_r}}, \qquad J(W) = -\sum_{i,k} y_{ik} \log p_k(\mathbf{x}_i).$$

Scores $(2, 1, 0)$ give approximately $(0.665, 0.245, 0.090)$.

Theorem 4.8: softmax calculus

With $\operatorname{LSE}(\mathbf{s}) = \log \sum_r e^{s_r}$ and $\mathbf{p} = \operatorname{softmax}(\mathbf{s})$:

(i) $\operatorname{softmax}(\mathbf{s} + c\mathbf{1}) = \operatorname{softmax}(\mathbf{s})$, $\operatorname{LSE}(\mathbf{s} + c\mathbf{1}) = \operatorname{LSE}(\mathbf{s}) + c$; stable evaluation $\operatorname{LSE}(\mathbf{s}) = M + \log \sum e^{s_r - M}$, $M = \max s_r$.

(ii) $\nabla \operatorname{LSE} = \mathbf{p}$; Jacobian $\partial p_k/\partial s_m = p_k(\delta_{km} - p_m)$, i.e. $\nabla^2 \operatorname{LSE} = \operatorname{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^\top \succeq 0$.

(iii) $\nabla_{\mathbf{s}}(-\log p_y) = \mathbf{p} - \mathbf{e}_y$; for the whole data set $\nabla_W J = (P - Y)^\top X$ (rows: classes).

(iv) $\log \operatorname{softmax}(\mathbf{s}) = \mathbf{s} - \operatorname{LSE}(\mathbf{s})\mathbf{1}$ and log-softmax is idempotent: $\log \operatorname{softmax}(\log \operatorname{softmax}(\mathbf{s})) = \log \operatorname{softmax}(\mathbf{s})$.

(v) $\max_k s_k \leq \operatorname{LSE}(\mathbf{s}) \leq \max_k s_k + \log Q$.

Proof. (i) e^c cancels.

(ii) Quotient rule; $\mathbf{v}^\top(\operatorname{diag}\mathbf{p} - \mathbf{p}\mathbf{p}^\top)\mathbf{v} = \sum p_k v_k^2 - (\sum p_k v_k)^2$ is the variance of v_K with $K \sim \mathbf{p}$.

(iii) $-\log p_y = -s_y + \operatorname{LSE}(\mathbf{s})$.

(iv) With $\boldsymbol{\ell} = \log \operatorname{softmax}(\mathbf{s})$, $e^{\ell_j} = p_j$ so $\operatorname{LSE}(\boldsymbol{\ell}) = \log \sum p_j = 0$.

(v) $e^M \leq \sum e^{s_r} \leq Qe^M$. ■

Adding the same vector to all rows of W leaves the softmax unchanged, so W is identifiable only up to such shifts.

5 Support Vector Machines

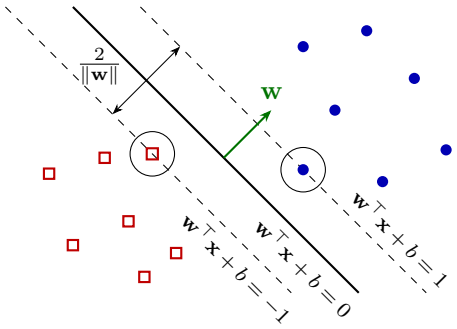
5.1 Geometry and the primal problem

Labels are $y_i \in \{\pm 1\}$, score $f(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b$. The distance from $\mathbf{x}_0$ to $\{\mathbf{w}^\top \mathbf{x} + b = 0\}$ is $|\mathbf{w}^\top \mathbf{x}_0 + b|/\|\mathbf{w}\|$ (move along the unit normal: $\mathbf{x}_0 - t\mathbf{w}/\|\mathbf{w}\|$ lies on the plane iff $t = (\mathbf{w}^\top \mathbf{x}_0 + b)/\|\mathbf{w}\|$). Rescaling $(\mathbf{w}, b)$ by $c > 0$ changes nothing, so for separable data normalize the closest functional margin to 1:

Definition 4.9: hard-margin SVM

$$\min_{\mathbf{w},b} \ \tfrac{1}{2}\|\mathbf{w}\|^2 \quad \text{s.t.} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1, \ i = 1, \ldots, n.$$

The support planes $\mathbf{w}^\top \mathbf{x} + b = \pm 1$ lie at distance $1/\|\mathbf{w}\|$ (the margin) from the boundary; the slab width is $2/\|\mathbf{w}\|$.



Linear SVM: the margin between the hyperplanes $\mathbf{w}^\top \mathbf{x} + b = \pm 1$ is $2/\|\mathbf{w}\|$; the circled points are support vectors.

5.2 Lagrangian, max–min inequality, weak duality

Theorem 4.10: max–min inequality

For any $F : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$, $\sup_x \inf_y F(x, y) \leq \inf_y \sup_x F(x, y)$. Equality can fail: for $F = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ (rows x, columns y) the left side is 0 and the right side 1.

Proof. For all x, y: $\inf_{y'} F(x, y') \leq F(x, y) \leq \sup_{x'} F(x', y)$. The left end does not depend on y and the right end not on x; take $\sup_x$ on the left, then $\inf_y$ on the right. ■

With $c_i = 1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b)$ and $\alpha_i \geq 0$, $L(\mathbf{w}, b, \boldsymbol{\alpha}) = \frac{1}{2}\|\mathbf{w}\|^2 + \sum_i \alpha_i c_i$. Since $\sup_{\alpha \geq 0} \alpha c = 0$ if $c \leq 0$ and $+\infty$ if $c > 0$, $\sup_{\boldsymbol{\alpha} \geq \mathbf{0}} L$ equals the objective on feasible points and $+\infty$ elsewhere; hence $p^\star = \inf_{\mathbf{w},b} \sup_{\boldsymbol{\alpha}} L$. The dual is $d^\star = \sup_{\boldsymbol{\alpha}} \inf_{\mathbf{w},b} L$ and the max–min inequality gives **weak duality** $d^\star \leq p^\star$: every dual value is a lower bound on the primal optimum. For the SVM (convex, strictly feasible) strong duality $d^\star = p^\star$ holds.

5.3 The dual problem

Theorem 4.11: SVM dual

$\nabla_{\mathbf{w}} L = \mathbf{0} \Rightarrow \mathbf{w} = \sum_i \alpha_i y_i \mathbf{x}_i$; finiteness of $\inf_b L$ requires $\sum_i \alpha_i y_i = 0$. The dual is

$$\max_{\boldsymbol{\alpha} \geq \mathbf{0}} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^\top \mathbf{x}_j \quad \text{s.t.} \quad \sum_i \alpha_i y_i = 0.$$

Complementary slackness $\alpha_i [1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b)] = 0$: $\alpha_i > 0$ only for support vectors, and $b = y_i - \mathbf{w}^\top \mathbf{x}_i$ for any such i.

Proof. b enters L linearly with coefficient $-\sum \alpha_i y_i$; if nonzero, $\inf_b = -\infty$. Substituting $\mathbf{w}$: $\frac{1}{2}\|\mathbf{w}\|^2 - \|\mathbf{w}\|^2 + \sum \alpha_i = \sum \alpha_i - \frac{1}{2}\|\mathbf{w}\|^2$. ■

Example 4.12: two points

$\mathbf{x}_1 = (0,0)$, $y_1 = -1$; $\mathbf{x}_2 = (2,2)$, $y_2 = +1$. The constraint forces $\alpha_1 = \alpha_2 = a$ and $\|\mathbf{x}_2\|^2 = 8$, so the dual is $2a - 4a^2$, maximized at $a = 1/4$. Then $\mathbf{w} = \frac{1}{4}(2,2) = (1/2, 1/2)$, $b = -1$ (from $\mathbf{x}_1$), the boundary is $x^{(1)} + x^{(2)} = 2$, $\|\mathbf{w}\| = 1/\sqrt{2}$, margin $\sqrt{2}$ (slab width $2\sqrt{2}$). Primal value $\frac{1}{2}\|\mathbf{w}\|^2 = 1/4$ equals the dual value $2a - 4a^2 = 1/4$.

Remark 4.13: soft margin and kernels

For non-separable data add slacks: $\min \frac{1}{2}\|\mathbf{w}\|^2 + C \sum \xi_i$ s.t. $y_i f(\mathbf{x}_i) \geq 1 - \xi_i$, $\xi_i \geq 0$, equivalently hinge loss $C \sum \max(0, 1 - y_i f(\mathbf{x}_i))$ plus ℓ_2. Its dual is identical except for the box constraint $0 \leq \alpha_i \leq C$. Replacing $\mathbf{x}_i^\top \mathbf{x}_j$ by a kernel gives the kernel SVM with score $\sum_i \alpha_i y_i k(\mathbf{x}_i, \mathbf{x}) + b$. Logistic regression fits a likelihood; the SVM maximizes a geometric margin.